

Sharing auxiliary roles without losing the finite-network rank deficit

Douglas Colkitt*

October 7, 2026

Abstract

We identify first-stage and third-stage side roles in the finite bit network used by OpenAI's integer-multiplication manuscript. An explicit matching makes the new frame transitions nested, removing one third of the side roles while preserving the absolute rank deficit. Combined with our previous routing and parameter audits, this supports the conditional exponent saving $\kappa = 2^{-75}$. The complex network is unchanged. The improvement concerns an asymptotic exponent; no practical runtime comparison is claimed.

1 Setting and the matching

We retain the finite-alphabet, fixed finite number of one-dimensional tapes model of *Integer multiplication below $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

All unchanged algorithmic interfaces remain assumptions. We use the previously audited $h = 46$ instance. Let $\mathcal{T}$ be the triples of $[h]$, $v = \binom{h}{3}$, $N = v^3$, $m = h^3$, and $z = 3\binom{h-3}{2}$. Two triples are neighbors when their intersection has size one. The rational label space is $F = \mathbb{Q}^h$ with form $I - J/9$; triple indicators have norm two and neighboring indicators are orthogonal.

Pair consecutive ground points. Define a permutation π of $\mathcal{T}$. If a triple contains a full pair and a singleton, keep the singleton and cycle its full pair among the pairs other than the singleton's pair. Otherwise it uses three different pairs: keep the point in the least indexed pair and flip the other two points to their partners. The first rule is a cycle for each fixed singleton; the second is an involution. Their classes are disjoint, and

$$|T \cap \pi(T)| = 1 \quad \text{for every } T \in \mathcal{T}.$$

Thus π is explicit and $t_T \perp t_{\pi(T)}$.

Fix a common ordering of triples, and let $n_S(j)$ be the j th neighbor of S . A stage-1 side role has fixed second and third triples (A, B) and physical first-factor target/source triples $(S, n_S(j))$. Pair it with the stage-3 side role with fixed first and second triples $(S, \pi(A))$ and third-factor target/source triples $(B, n_B(j))$. This is bijective: recover A with π^{-1} and j from the final neighbor list. There are $R = Nz$ such pairs. Only these side roles are shared; stage-2 side roles and all central roles remain separate.

*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.

2 Scalar correctness and frame accounting

Every first-stage invocation finishes before the third stage starts. A scalar invocation restores all its auxiliary values, regardless of their initial values. Identifying the paired roles therefore preserves the scalar bank exchange and restoration of every remaining auxiliary value. No simultaneous roles within an invocation are identified.

Scalar restoration alone does not prove an improvement in the rank count. The last stage-1 gate on the shared role has label

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad \dim E = h.$$

The first stage-3 gate has label

$$H = \langle t_S \otimes t_{\pi(A)} \rangle_{F \otimes F}^\perp \otimes \langle t_B \rangle, \quad \dim H = h^2 - 1.$$

For any $x \in F$, the inner product of $x \otimes t_A$ with $t_S \otimes t_{\pi(A)}$ vanishes. Hence $E \subset H$. Both labels are nondegenerate under the source's tensor decompositions.

Proposition 1 (Exact saving per shared role). *The identification removes one role and exactly m units of total rational edge rank, while preserving all gate compatibility and endpoint conditions.*

Proof. The old terminal edges are $E \rightarrow F^{\otimes 3}$ and $0 \rightarrow H$. Their total projection-difference rank is $(m - h) + (h^2 - 1)$. The new edge $E \rightarrow H$ has rank $h^2 - 1 - h$, exactly m less. All other edges and gate labels remain unchanged. The shared role still has initial matrix zero and final matrix I_m and is fixed by the completed scalar permutation. All data endpoints are unchanged. Thus the identity $M_{\text{sink}(\rho(w))} - M_{\text{source}(w)} = I_m$ holds on every role. The common-frame and rational-shear arguments apply to this fixed circuit. $\square$

Consequently

$$\begin{aligned} R &= 9475984020888000, \\ W_b^* &= W_b - R = 18958995769111200, \\ s_b^* &= s_b - mR = 1845392811609813681600, \\ W_b^*m - s_b^* &= N - 2L_b > 0. \end{aligned}$$

The relative deficit increases because the absolute deficit is preserved while the denominator decreases. The resulting new relative deficit is

$$\eta_b^* = \frac{9}{29015910268}.$$

This changes a network hypothesis of the previous fixed-network tuning bound; it does not contradict that bound.

3 Conditional multiplication witness

Put

$$a = \frac{27}{10^{12}}, \quad a_c = \frac{9}{500000000000}, \quad L = \frac{5743}{500}.$$

The existing exact logarithm certificate proves $\log m < L$, and exact arithmetic proves $\eta_b^* > aL$. Therefore $m^{1-a} > m(1 - aL) > m(1 - \eta_b^*)$. The unchanged complex network supports $\sigma = 1 - a_c$ as before. Choose

$$\begin{aligned} \tau &= 1 - a, \quad \sigma = 1 - a_c, \quad \epsilon = \frac{1}{21}, \quad \beta = \frac{9}{10}, \quad c = \frac{9a}{10}, \\ \lambda &= 1 - \frac{19a^2}{20}, \quad \lambda' = 1 - \frac{9a^2}{10}, \quad \delta = \frac{1}{16}, \quad C_1 = 20. \end{aligned}$$

The packed condition is $\tau(1 + c/\beta) = 1 - a^2 < \lambda < \lambda' < 1$. The separate conditions involving σ follow from $19a^2/20 < a_c$ and $9a^2 < a_c$. The seven margins are

[illegible]

with $g_1 > 1/2$, $g_4 = 20a/21$, $g_5 = 13/112$, $g_6 = 299/336$ and $g_7 = 1/21$, all larger than G . For $\rho = G - 2^{-75} > 0$, the existing strict-gap argument absorbs fixed logarithmic factors and yields $O(n(\log n)^{1-2^{-75}})$ conditionally.

The dimension exponent, stopping threshold and guard bound match the 2^{-76} tuning note. Thus the guard is $p^{20/21} = o(p)$, $\alpha = \Theta(p^{23/84})$ and $\gamma = O(p^{25/42}) = o(p)$. The enlarged chunk has $K = \Theta(p^{3a/70})$, still $K = o(\ell)$ and $K/\log p \rightarrow \infty$, where $\ell = \Theta(p^{20/21})$. Record length remains superpolynomial. The bit-network change affects only finite constants in its fixed-tape implementation; the complex network, its arithmetic guard analysis, the numerical maps and final rounding are retained. No appeal is made to the separate recurrence refinement, which had assumed $\sigma \leq \tau$; here $\sigma > \tau$ and the original layer inequalities pass.

4 Artifacts and review boundary

The companion repository supplies an exact parameter and count certificate, a standalone patch, and finite checks:

- `scripts/reuse_network.py` and `certificates/stage-reuse.json`;
- `patches/h46-shared-side-75.patch`;
- `tests/test_reuse_network.py`.

The matching is checked on every one of the 15,180 triples at $h = 46$. Small rational projection checks verify the orthogonality used in the nesting argument. A complete $h = 6$ scalar model checks the bank exchange and restoration of 151,200 auxiliary roles with nonzero initial values. These finite tests support the written proof; they do not independently establish the upstream algorithm or its time bounds. No global optimality claim is made.